

A711

$\theta > 0$

Q1]  $X_1 \dots X_n \sim f(x, \theta) = \sigma x^{\sigma-1} \mathbb{1}_{(0,1)}(x)$   
 $= \sigma \frac{x^\sigma}{x}$

$\sigma X = Y$   
 $X = \frac{Y}{\sigma} \quad \left( \frac{dx}{dy} \right)$   
 ~~$\frac{dx}{dy} = \frac{1}{\sigma}$~~   
 $X = \frac{Y}{\sigma}$   
 $\frac{dx}{dy} = \frac{1}{\sigma}$

~~$f_Y = \sigma \frac{x^\sigma}{x} \cdot \frac{1}{\sigma}$~~

$X^\sigma = Y \quad \mathbb{1}_{(0,1)}(x)$   
 $\sigma \ln(x) = \ln(Y)$   
 $\ln(Y) = \sigma \ln(x)$   
 $X = Y^{1/\sigma}$

$\frac{dx}{dy} = \frac{1}{\sigma} Y^{\frac{1}{\sigma}-1} = \frac{1}{\sigma} Y^{\frac{1-\sigma}{\sigma}}$

$f_Y(y) = \cancel{\sigma} \cdot \frac{1}{\sigma} \cdot Y^{\frac{1-\sigma}{\sigma}} \cdot \frac{Y}{Y^{1/\sigma}} = Y^{\frac{1-\sigma}{\sigma}} \cdot Y^{1-1/\sigma}$   
 $= Y^{\frac{1-\sigma}{\sigma} - 1 + 1 - 1/\sigma} = 1 \cdot \mathbb{1}_{(0,1)}(y)$   
 $f_Y = 1 \cdot \mathbb{1}_{(0,1)}(y)$

$\ln(x) = y$

$x = e^y$

$\frac{dx}{dy} = e^y$

$f_Y(y) = e^y \cdot \sigma e^{y\sigma - y}$

$= \sigma e^{y\sigma}$

$Y \sim \exp(\sigma)$

$\ell(x|\theta) = \ln(\sigma) + (\sigma-1) \ln(x)$   
 $\ell(\bar{X}|\theta) = n \ln(\sigma) + (\sigma-1) \sum \ln(x_i)$   
 $\frac{d\ell}{d\sigma} = \frac{n}{\sigma} + \sum \ln(x_i) = 0$   
 $-\frac{n}{\sigma^2} = 0$

$Y \sim U_{ref}$

$$L(X) = Y_1$$

$$X = e^{Y_1}$$

$$\frac{dY}{dX} = e^{Y_1}$$

$$f_Y(y) = e^{y_1} \sigma e^{y_0 - 1}$$

$$= \sigma e^{y_0} \rightarrow Y \sim \exp(\sigma)$$

$$Y_1 + \dots + Y_n \sim \text{Gam}(n, \sigma)$$

$$2\sigma(Y_1 + \dots + Y_n) \sim \chi^2_{2n}$$

$$Q = 2\sigma \sum \log(x_i) \sim \tilde{\chi}^2_{(2n)}$$

$$a) P_r(a \leq Q \leq b) = 0.9$$

$$P_r(Q \leq a) = 0.05 \rightarrow a = \tilde{\chi}^2_{0.05(2n)}$$

$$P_r(Q \geq b) = 0.05 \rightarrow b = \tilde{\chi}^2_{0.95(2n)}$$

$$P_r\left(\tilde{\chi}^2_{0.05(2n)} \leq 2\sigma \sum \log(x_i) \leq \tilde{\chi}^2_{0.95(2n)}\right) = 0.9$$

$$P_r\left(\frac{\tilde{\chi}^2_{0.05(2n)}}{2\sum \log(x_i)} \leq \sigma \leq \frac{\tilde{\chi}^2_{0.95(2n)}}{2\sum \log(x_i)}\right) = 0.9$$

$$\tilde{I}_{0.90, \sigma}(\sigma)$$

$$b) P_r(Q \leq b) = 0.99 \rightarrow P_r(2\sigma \sum \log(x_i) \leq \tilde{\chi}^2_{0.99(2n)}) = 0.99$$

$$P_r\left(\sigma \leq \frac{\tilde{\chi}^2_{0.99(2n)}}{2\sum \log(x_i)}\right) = 0.99$$

$$P_r(a \leq Q) = 0.99 \rightarrow P_r(\tilde{\chi}^2_{0.01(2n)} \leq 2\sigma \sum \log(x_i))$$

P2]  $X_1, \dots, X_n \sim \text{Pareto}(\sigma)$  B30

$$f(x|\sigma) = \frac{x}{\sigma^2} \exp\left\{-\frac{x^2}{2\sigma^2}\right\} \mathbb{I}_{(0,\infty)}(x)$$

$$L(x|\sigma) = \prod_{i=1}^n \frac{x_i}{\sigma^2} \exp\left\{-\frac{x_i^2}{2\sigma^2}\right\} \quad \sigma^{-2}$$

$$\ell(x|\sigma) = \sum \ln(x_i) - n \ln(\sigma^2) - \frac{\sum x_i^2}{2\sigma^2} = 0$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{\sigma^2} + \frac{\sum x_i^2}{2(\sigma^2)^2} = 0$$

$$\sigma^2 = \left( \frac{\sum x_i^2}{2n} \right)^2$$

$$\ell(x|\sigma) = \sum \ln(x_i) - n \ln(\sigma^2) - \frac{\sum x_i^2}{2\sigma^2}$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{\sum x_i^2}{2\sigma^2} = 0$$

$$V_L(X_K) = 2K$$

$$\hat{\sigma}^2 = \frac{\sum x_i^2}{2N}$$

$$\mathbb{E}(x_i) = \sigma \sqrt{\pi/2}$$

$$V_L(x_i) = \frac{4-\pi}{2} \sigma^2$$

$$\mathbb{E}(\hat{\sigma}^2) = \frac{\sum \mathbb{E}(x_i^2)}{2N}$$

$$= \frac{\sum 2\sigma^2}{2n} = \sigma^2$$

$$\mathbb{E}(x_i^2) = V_L(x_i) + \mathbb{E}(x_i)^2$$

$$= \frac{4-\pi}{2} \sigma^2 + \sigma^2 \frac{\pi}{2}$$

$$= 2\sigma^2$$

$$ECM(\hat{\sigma}^2) = V_L(\hat{\sigma}^2) = \frac{\sum V_L(x_i^2)}{4n^2}$$

$$V_L(x_i^2) = \mathbb{E}(x_i^4) - \mathbb{E}(x_i^2)^2$$

$$= 8\sigma^4 - 4\sigma^4$$

$$V_L(x_i) = 4\sigma^4$$

$$ECM(\hat{\sigma}^2) = \frac{\sigma^4}{n}$$

$$\hat{\sigma}^2 \pm 2\sqrt{EMC(\hat{\sigma}^2)}$$

$$\sigma^2 \in \left\{ \frac{\sum_{i=1}^n X_i^2}{2n} \pm \frac{2\sigma^2}{\sqrt{n}} \right\}$$

$$P\left(\frac{\sum_{i=1}^n X_i^2}{2n} \pm \frac{2\sigma^2}{\sqrt{n}} \leq \sigma^2 \leq \frac{\sum_{i=1}^n X_i^2}{2n} + \frac{2\sigma^2}{\sqrt{n}}\right)$$

$$-\frac{2\sigma^2}{\sqrt{n}} \leq \sigma^2 - \frac{\sum_{i=1}^n X_i^2}{2n} \leq \frac{2\sigma^2}{\sqrt{n}}$$

$$\sigma^2 \left(1 - \frac{2}{\sqrt{n}}\right) \leq \sigma^2 \leq \sigma^2 \left(1 + \frac{2}{\sqrt{n}}\right)$$

$$\sum_{i=1}^n Z_i^2 \sim \chi_{2n}^2$$

$$2n\left(1 - \frac{2}{\sqrt{n}}\right) \leq \frac{\sum_{i=1}^n X_i^2}{\sigma^2} \leq 2n\left(1 + \frac{2}{\sqrt{n}}\right)$$

$$\leq \sum_{i=1}^n Y_i^2 \leq 2n\left(1 + \frac{2}{\sqrt{n}}\right) \sim$$

$$f_Y(y) = \frac{\sigma \cdot \sigma Y_i}{\sigma^2} \exp\left\{-\frac{\sigma^2 Y_i^2}{\sigma^2}\right\}$$

$$\frac{X_i^2}{\sigma^2} = Y_i$$

$$X_i = \sigma Y_i$$

$$\frac{dx}{dy} = \sigma$$

$$= Y_i \exp\{-Y_i^2\} \sim \text{Rayler}(1) \sim \text{Exp}(1) \sim \text{Gam}(1, 1)$$

$$\sum_{i=1}^n Y_i^2 \sim \text{Gam}(n, 1) \sim \chi_{2n}^2$$

$$Y_i^2 = Z_i$$

$$Y_i = \sqrt{Z_i}$$

$$\frac{\partial y}{\partial z} = \frac{1}{2\sqrt{z_i}}$$

$$f_Z(z) = \frac{1}{2\sqrt{z_i}} \exp\left\{-\sqrt{z_i}\right\} = \frac{1}{2} \exp\{-z_i\} \sim \text{Exp}(1) \sim \chi_2^2$$

$$P_r(2n - 4\sqrt{n} \leq \chi^2(2n) \leq 2n + 4\sqrt{n})$$

| $n =$ | $P(D)$ |
|-------|--------|
| 5     | 0,9588 |
| 10    | 0,9585 |
| 100   | 0,955  |

$$b) Q = \frac{1}{\sigma^2} \sum x_i^2 \sim \chi^2(2n)$$

$$P_r(a \leq Q \leq b) = 1 - \alpha$$

$$P_r\left(\chi^2_{\alpha/2, 2n} \leq \frac{1}{\sigma^2} \sum x_i^2 \leq \chi^2_{1-\alpha/2, 2n}\right) = 1 - \alpha$$

$$\left( \frac{\sum x_i^2}{\chi^2_{1-\alpha/2, 2n}} \leq \sigma^2 \leq \frac{\sum x_i^2}{\chi^2_{\alpha/2, 2n}} \right)$$

$$c) n=10 \quad \sum x_i^2 = 21,447 \rightarrow \hat{\sigma}^2 = \frac{21,447}{2 \cdot 10} = 1,072$$

$$\text{Interv. w. a} \quad (\hat{\sigma}^2 \pm \text{EMC}(\hat{\sigma}^2)) = \left( 1,072 - \frac{2}{\sqrt{10}}, 1,072 + \frac{2}{\sqrt{10}} \right)$$

$$= (0,439, 1,404) \quad L=1,265 \quad \text{Coverage} = 0,9585$$

$$\text{Interv. w. b) } \left( \frac{21,447}{34,17}, \frac{21,447}{9,59} \right) = (0,628, 2,236) \quad L=1,609 \quad C=0,95$$

$$P3) \quad X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2)$$

$$Y_1, \dots, Y_n \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2)$$

$$\bar{X} \sim N(\mu_1, \sigma_1^2/n)$$

$$\bar{Y} \sim N(\mu_2, \sigma_2^2/n)$$

$$\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \sigma_1^2/n + \sigma_2^2/n) \quad -2\text{Cov}(\bar{X}, \bar{Y})$$

$$\bar{X} - \bar{Y} - (\mu_1 - \mu_2) \sim N(0, \sigma_1^2/n + \sigma_2^2/n)$$

$$\sqrt{n} \frac{(\bar{X} - \bar{Y} - (\mu_1 - \mu_2))}{\sqrt{\sigma_1^2 + \sigma_2^2}} \sim N(0, 1)$$

$$\bar{X} - \bar{Y} - (\mu_1 - \mu_2)$$

$$Z_i = X_i - Y_i \sim N\left(\frac{\mu_1 - \mu_2}{\mu_Z}, \frac{\sigma_1^2 + \sigma_2^2 - 2\text{Cov}(X_i, Y_i)}{\sigma_Z^2}\right)$$

$$Q = \frac{\sqrt{n}(\bar{Z} - \mu_Z)}{\sigma_Z} \sim N(0, 1)$$

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n (X_i - Y_i)$$

$$S_Z^2 = \frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})^2$$

$$Q = \frac{\sqrt{n}(\bar{Z} - \mu_Z)}{S_Z} \sim \text{t}_{(n-1)}$$

Per similitudine di F e t

$$P\left(\tau_{\alpha/2}^{(n-1)} \leq Q \leq \tau_{1-\alpha/2}^{(n-1)}\right) = 1-\alpha$$

$$P\left(\bar{Z} - \tau_{\alpha/2}^{(n-1)} \frac{S_Z}{\sqrt{n}} \leq \mu_Z \leq \bar{Z} + \tau_{1-\alpha/2}^{(n-1)} \frac{S_Z}{\sqrt{n}}\right) = 1-\alpha$$

P4  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$

$$Y_1 \dots Y_m \stackrel{\text{ind}}{\sim} \exp(\lambda_2)$$

$$\sum_i x_i \sim \text{Geom}(\lambda, \lambda)$$

$$2 \lambda_1 \sum_i^N x_i \sim \text{Ca}(n, 1/2) \sim \tilde{\chi}_{(2n)}^2$$

$$2\lambda_2 \sum_1^m \gamma_i \sim G_{ca}(m, 1/2) \sim \chi^2_{(2m)}$$

$$\frac{\lambda_1 \bar{x}}{\lambda_2 \bar{y}} = Q = \frac{\int \lambda_1 \tilde{x}^{1/2m}}{\int \lambda_2 \tilde{y}^{1/2m}} \sim F_{(2m, 2m)} = \frac{\tilde{\chi}_k/k}{\chi_e^2/l}$$

$$Pr(a \leq Q \leq b) = 1 - 2$$

$$R_1(F_{2/2}(2m, 2m)) \leq \frac{\lambda_1}{\lambda_2} \frac{\bar{X}}{\bar{Y}} \leq F_{1-\alpha/2}(2m, 2m) = 1 - \alpha$$

$$F_{\alpha/2}(z_m, z_m) \frac{\bar{Y}}{\bar{X}} \leq \frac{\lambda_1}{\lambda_2} \leq \frac{\bar{Y}}{\bar{X}} F_{1-\alpha/2}(z_m, z_m)$$

$$I_C\left(\frac{\lambda}{\lambda_r}\right) = \left( F_{\lambda_r}(2m, 2m) \frac{\overline{Y}}{\overline{X}}, F_{\lambda_r}(2m, 2m) \frac{\overline{Y}}{\overline{X}} \right)$$

Andhra Pradesh

$$I_c ($$

$$\sqrt{n}(\bar{X} - 1/\lambda_1) \rightarrow N(0, 1/\lambda_1^2)$$

$$\gamma(T) = \frac{1}{T} \rightarrow \gamma'(T) = -\frac{1}{T^2}$$

$$\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathcal{N}(C, \Sigma)$$

$$\sqrt{m} \left( \frac{1}{\sqrt{y}} - \lambda_n \right) \rightarrow N(0, \lambda_n^2)$$

$$\frac{1}{\lambda} \rightarrow N(\lambda_1, \frac{\lambda_1^2}{n})$$

$$\frac{1}{\sqrt{n}} \rightarrow N\left(\lambda_1, \frac{\lambda_1^2}{n}\right)$$

$$\gamma(t) \approx \gamma(\sigma) + \gamma'(\sigma)(t - \sigma)$$

$$\#(\tau(\tau)) = \tau(0)$$

$$V_{\omega}(\gamma(\tau)) = [\gamma'(0)]^2 V_{\omega}(\tau)$$

*[Handwritten signature]*

$$\lambda_1^4 \cdot \frac{1}{\lambda_1^2} = \lambda_1^2$$

$$\frac{14}{Q} = \left( \frac{\frac{1}{\bar{X}} - \frac{1}{\bar{Y}} - (\lambda_1 - \lambda_2)}{\sqrt{\frac{\lambda_1^2}{n} + \frac{\lambda_2^2}{m}}} \right) \sim N(0,1)$$

$$I_C(\lambda_1 - \lambda_2) = \left( \frac{1}{\bar{X}} - \frac{1}{\bar{Y}} \mp z_{1-\alpha/2} \sqrt{\frac{\lambda_1^2}{n} + \frac{\lambda_2^2}{m}} \right)$$

Intérêt nul dépend de  $\lambda_1$  et  $\lambda_2$   
 Utilisation propriétés asymptotiques (Théorème de Slutsky)  
 On peut remplacer  $\lambda_1$  par  $\frac{1}{\bar{X}}$  et  $\lambda_2$  par  $\frac{1}{\bar{Y}}$

$$I_C(\lambda_1 - \lambda_2) = \left( \frac{1}{\bar{X}} - \frac{1}{\bar{Y}} \mp z_{1-\alpha/2} \sqrt{\frac{(1/\bar{X})^2}{n} + \frac{(1/\bar{Y})^2}{m}} \right)$$